

Aerial Human Action Recognition using Hybrid YOLO–CLIP Architecture with SAHI

Abstract—Human action recognition from aerial drone imagery presents substantial challenges due to extreme top-down viewpoints, severe scale variation, motion blur, and dense background clutter. In high-altitude surveillance footage, human subjects often occupy only a few pixels, causing conventional end-to-end convolutional and video-based action recognition models to fail after standard image resizing. Although recent aerial action recognition approaches improve performance through deeper backbones or temporal modeling, they remain highly sensitive to resolution loss, require large annotated video sequences, and struggle with visually ambiguous actions observed from overhead perspectives.

Moreover, existing methods tightly couple human detection and action classification within a single pipeline, leading to compounding errors and limited generalization in real-world drone surveillance scenarios. The absence of explicit mechanisms for small-object preservation further exacerbates detection failures, while pose- or keypoint-based approaches degrade rapidly at altitudes above 20 meters.

In this paper, we propose a hybrid, two-stage aerial human action recognition framework that explicitly decouples mechanical localization from semantic action understanding. The proposed approach integrates YOLOv11 with Slicing Aided Hyper Inference (SAHI) to preserve spatial resolution and reliably detect small-scale humans through overlapping high-resolution inference. Detected regions are subsequently classified using a frozen Contrastive Language–Image Pre-training (CLIP) visual encoder augmented with a lightweight trainable adapter, enabling robust semantic discrimination of actions without end-to-end retraining.

The framework operates in a training-efficient manner, leveraging pretrained models exclusively while maintaining strong adaptability to aerial viewpoints. We evaluate our method on the challenging *Aeriform In-Action* dataset, consisting of 32 high-resolution drone videos and 13 action categories captured under real-world conditions. Experimental results demonstrate a significant improvement in recognition performance, achieving an overall accuracy of 94.53%, substantially outperforming conventional CNN-based baselines. These findings validate that separating small-object detection from semantic reasoning provides a scalable and robust solution for aerial human action recognition in surveillance and search-and-rescue applications.

Index Terms—Aerial action recognition, YOLO, CLIP, SAHI, small object detection, multimodal learning

I. INTRODUCTION

Automatic recognition of human actions is a fundamental problem in computer vision and a critical component of modern video surveillance systems. Accurate interpretation of human behavior enables a wide range of applications, including gait recognition, group activity analysis, suspicious event detection, and gesture-based interaction systems [1]–[4]. Traditionally, human action recognition has been studied extensively in ground-based camera settings, where subjects

are captured at close range with stable viewpoints and relatively high spatial resolution. In such scenarios, deep learning models can reliably extract appearance, motion, and pose-based features to classify actions.

However, conventional surveillance systems relying on fixed cameras suffer from limited coverage, rigid viewpoints, and restricted deployment flexibility. To overcome these limitations, Unmanned Aerial Vehicles (UAVs), commonly referred to as drones, have emerged as a powerful alternative for large-scale surveillance, disaster response, and search-and-rescue operations [1], [6]. Drones provide dynamic viewpoints, rapid deployment, and wide-area monitoring capabilities that are infeasible for static camera infrastructures. As a result, aerial human action recognition has become an increasingly important yet challenging research direction.

Despite its potential, recognizing human actions from aerial footage remains considerably more difficult than its ground-based counterpart. Human actions can generally be categorized into atomic actions, such as walking, running, sitting, and standing, and interaction-based actions, which include human–human interactions (e.g., hugging, handshaking) and human–object interactions (e.g., carrying, reading). While these categories are relatively distinguishable in frontal or side-view imagery, they become highly ambiguous when observed from top-down aerial perspectives. The challenge is further exacerbated by the fact that drone-mounted cameras typically operate at altitudes ranging from 20 to 30 meters, causing human subjects to occupy only a few pixels in high-resolution frames.

In aerial surveillance scenarios, raw 4K video frames must often be resized to standard input resolutions (e.g., 224×224 or 640×640) to fit deep neural network architectures. This resizing process severely degrades fine-grained visual cues, frequently reducing human figures to fewer than five pixels in height, effectively collapsing discriminative information into noise. Consequently, standard convolutional neural networks and video-based action recognition models fail to generalize effectively in aerial contexts. Furthermore, aerial footage is affected by additional real-world challenges such as camera motion, illumination variation, occlusion among multiple actors, cluttered backgrounds, and continuously changing viewpoints.

Several prior works have attempted to address these challenges by adapting ground-based architectures to aerial data or by introducing deeper backbones and temporal modeling techniques. However, such approaches often require large amounts of annotated video data, incur significant computational overhead, and remain sensitive to resolution loss. Pose-

and keypoint-based methods, which have shown promise in ground-level action recognition, also degrade rapidly in aerial imagery due to poor keypoint visibility at high altitudes. Additionally, many existing pipelines tightly couple human detection and action recognition into a single end-to-end model, causing detection errors to propagate directly into the classification stage and limiting robustness in unconstrained environments.

Motivated by these limitations, this work proposes a fundamentally different design philosophy for aerial human action recognition—one that explicitly separates the mechanical task of localizing humans from the semantic task of understanding their actions. Building upon the insights of prior dataset-centric studies while addressing their architectural constraints, we introduce a hybrid two-stage framework that integrates three complementary technologies: YOLOv11 for object detection, Slicing Aided Hyper Inference (SAHI) for small-object preservation, and Contrastive Language–Image Pre-training (CLIP) for semantic action understanding.

In the first stage, SAHI is employed as a resolution-preserving inference mechanism that slices high-resolution aerial frames into overlapping patches. This approach acts as a magnification strategy, allowing YOLOv11 to detect small-scale humans without the destructive effects of global image resizing. By performing detection on localized high-resolution regions and subsequently merging predictions using global non-maximum suppression, the system significantly improves human localization accuracy in dense aerial scenes.

In the second stage, the detected human regions are passed to a CLIP-based semantic classification module. Instead of training a conventional CNN from scratch, we leverage the rich multimodal representations learned by CLIP from large-scale image–text pretraining. A lightweight trainable adapter is appended to the frozen CLIP visual encoder, enabling efficient adaptation to aerial action categories while preserving generalized semantic knowledge. This design allows the model to disambiguate visually similar actions—such as standing and walking—by relying on high-level semantic cues rather than brittle low-level appearance features alone.

The proposed framework is training-efficient, modular, and robust to the unique challenges of aerial surveillance. It avoids the heavy computational cost of 3D convolutions and recurrent architectures, requires significantly less labeled data, and generalizes effectively across varying viewpoints and environmental conditions. We evaluate our approach on the challenging *Aeriform In-Action* dataset, which consists of high-resolution aerial videos capturing diverse atomic and interaction-based actions in real-world settings. Experimental results demonstrate that our hybrid architecture substantially outperforms conventional CNN-based baselines, achieving an overall recognition accuracy of 94.53%.

II. RELATED WORK

Human action recognition in aerial imagery lies at the intersection of small object detection, viewpoint-robust visual understanding, and efficient temporal modeling. While significant

progress has been made in each of these areas independently, their integration for real-world aerial surveillance remains an open challenge.

A. Small Object Detection in Aerial Imagery

Standard object detection frameworks, including single-stage detectors such as YOLO and two-stage detectors such as Faster R-CNN, are typically designed to operate on resized inputs (e.g., 416×416 or 640×640). In aerial surveillance scenarios, this resizing process leads to severe information loss, as human subjects captured from high altitudes often occupy only a few pixels in the original high-resolution frame. Consequently, small-scale humans are frequently compressed into indistinguishable pixel noise, a phenomenon commonly referred to as the *small object problem* [7].

Several approaches have attempted to mitigate this issue through feature pyramid networks, multi-scale training, or anchor refinement strategies. However, these techniques provide only marginal improvements when object sizes fall below a critical resolution threshold. More recently, Slicing Aided Hyper Inference (SAHI) has been proposed as a resolution-preserving inference strategy that divides high-resolution images into overlapping patches and performs detection independently on each slice [8]. This approach has demonstrated significant gains in detecting small objects in UAV imagery and agricultural monitoring, motivating its adoption in aerial human detection pipelines.

B. Aerial Human Action Recognition

Action recognition has traditionally been dominated by video-based models such as 3D Convolutional Neural Networks and recurrent architectures including LSTMs and GRUs [2]. These models rely heavily on temporal motion cues and dense frame sequences to distinguish between actions. While effective in ground-based datasets, such approaches face critical limitations in aerial contexts. First, they are computationally expensive, making them impractical for deployment on GPU-constrained drone edge devices or for processing long-duration surveillance streams. Second, their performance degrades substantially when temporal motion cues are weak or visually ambiguous due to top-down viewpoints.

Frame-based action recognition methods offer a computationally lighter alternative, but they struggle with fine-grained action discrimination in aerial views. Actions such as standing and walking often appear geometrically similar from overhead perspectives, and the absence of explicit temporal modeling further exacerbates this ambiguity. Moreover, pose- and keypoint-based approaches fail at typical drone altitudes due to poor joint visibility and self-occlusion.

C. Semantic and Multimodal Representations

Recent advances in vision–language models have shown that large-scale multimodal pretraining can produce semantically rich visual representations that generalize across tasks and domains. Models such as CLIP learn aligned image–text

embeddings from internet-scale datasets, enabling strong zero-shot and few-shot performance on downstream classification tasks [5]. Unlike conventional CNNs trained solely on visual labels, these models encode high-level semantic relationships that are particularly valuable when low-level appearance cues are unreliable.

However, the application of multimodal representations to aerial human action recognition remains underexplored. Most existing works treat action recognition as a purely visual classification problem and fail to leverage semantic context that could help disambiguate visually similar actions observed from aerial viewpoints.

D. Challenges in Aerial Action Recognition

Despite ongoing progress, several key challenges persist in aerial human action recognition:

- **Small Object Degradation:** High-resolution drone footage must be resized for standard detectors, causing small human subjects to vanish into pixel-level noise and severely degrading detection reliability.
- **Viewpoint-Induced Semantic Ambiguity:** Top-down aerial angles suppress critical visual cues such as limb articulation and posture, making it difficult to distinguish between subtly different actions (e.g., standing versus walking) using frame-based methods alone.
- **Temporal Modeling vs. Efficiency Trade-off:** While temporal models improve action discrimination, they introduce substantial computational overhead and latency, limiting their applicability for real-time or edge-based deployment.
- **Error Propagation in End-to-End Pipelines:** Many existing systems tightly couple detection and action classification, allowing localization errors to cascade into recognition failures and reducing robustness in unconstrained environments.

These limitations highlight the need for architectures that preserve spatial resolution for small-object detection, leverage semantic reasoning to compensate for viewpoint ambiguity, and maintain computational efficiency. Motivated by these challenges, our work adopts a hybrid two-stage design that decouples human localization from semantic action understanding, integrating SAHI-based detection with CLIP-driven semantic classification to address the core shortcomings of prior approaches.

III. PROPOSED METHOD

This section presents the proposed hybrid framework for aerial human action recognition. Inspired by prior two-stage evaluation pipelines for aerial surveillance, the proposed system adopts a modular design that integrates human detection and action recognition while explicitly addressing the challenges of small object scale and semantic ambiguity in aerial imagery. Unlike end-to-end approaches, the framework decouples localization and classification to improve robustness and scalability in real-world drone-based surveillance scenarios.

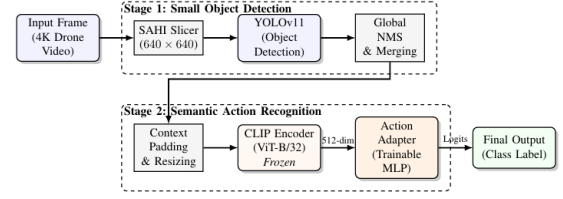


Fig. 1. Proposed Architecture. The pipeline consists of two stages: (1) Slicing Aided Hyper Inference (SAHI) with YOLOv11 for detecting small-scale persons, and (2) A CLIP-based Adapter network for classifying actions from the detected crops.

Fig. 1: Proposed hybrid architecture for aerial human action recognition. The pipeline consists of two stages: (1) Slicing Aided Hyper Inference (SAHI) with YOLOv11 for detecting small-scale human subjects from high-resolution drone frames, and (2) a CLIP-based semantic action recognition module with a trainable adapter for classifying actions from the detected crops.

A. Overall Architecture

The proposed system follows a two-stage processing pipeline consisting of a human detection module and a semantic action recognition module. Given an input aerial video, frames are first extracted and processed independently. In the first stage, human instances are localized using a YOLOv11-based detector enhanced with Slicing Aided Hyper Inference (SAHI) to preserve spatial resolution. In the second stage, the detected human regions are passed to a CLIP-based action recognition module that semantically classifies the actions being performed.

This separation effectively divides the visual understanding task into two complementary components: (i) locating small-scale human subjects (the “Eye” of the system), and (ii) interpreting their actions using high-level semantic representations (the “Brain” of the system). A high-level overview of the workflow is illustrated in Fig. 2.

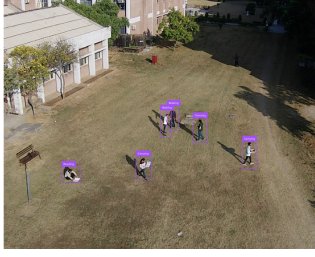
Fig. 1 illustrates the overall architecture of the proposed framework, highlighting the separation between small-object detection and semantic action recognition.

B. YOLOv11-based Person Detection

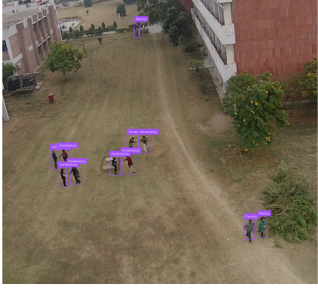
For human localization, we employ YOLOv11, a single-stage object detector optimized for both accuracy and computational efficiency. YOLOv11 formulates object detection as a regression problem, predicting bounding box coordinates and confidence scores directly from the input image. The image is divided into an $S \times S$ grid, where each grid cell is responsible for detecting objects whose centers fall within it.

In this work, the detector is trained to identify a single object category, namely humans, which simplifies the detection task and reduces unnecessary classification overhead. YOLOv11 was selected due to its favorable trade-off between detection accuracy and inference speed, making it suitable for aerial surveillance pipelines where real-time or near real-time processing is desirable.

However, when applied directly to resized aerial frames, even high-capacity detectors suffer from significant performance degradation due to the extremely small size of human



(a) YOLOv11-based detection of individual actions in aerial scenes.



(b) Detection of interaction-based actions under complex aerial conditions.

Fig. 2: Qualitative results of YOLOv11-based person detection in aerial imagery. The detector successfully localizes multiple small-scale human subjects and assigns preliminary action labels under varying spatial layouts, illumination conditions, and crowd densities. These detections serve as inputs to the subsequent SAHI-based slicing and semantic action classification stages.

subjects. To address this limitation, YOLOv11 is integrated with SAHI, as described in the following subsection.

C. SAHI-based Slicing Strategy

To mitigate the small object problem inherent in aerial imagery, we adopt a tiled inference strategy using Slicing Aided Hyper Inference (SAHI). Instead of resizing the entire high-resolution frame to a fixed input size, SAHI divides each frame into overlapping slices and performs detection independently on each slice.

1) *Inference Logic*: Given an input frame of resolution 3840×2160 , the image is partitioned into sliding windows (slices) of size $M \times N$, with an overlap ratio O between adjacent slices. Detection is performed k times, once per slice, using the YOLOv11 detector. This strategy ensures that small human instances occupy a larger proportion of each slice, thereby improving detection confidence and localization accuracy.

The detections from all slices are subsequently merged using a global Non-Maximum Suppression (NMS) operation to eliminate duplicate predictions arising from overlapping regions.

2) *Coordinate Mapping*: To ensure accurate placement of detected humans within the original frame, local bounding box coordinates predicted on individual slices are mapped back

to the global image coordinate space. If a detection occurs at local coordinates (x_{loc}, y_{loc}) within a slice whose top-left offset in the original image is (O_x, O_y) , the corresponding global coordinates are computed as:

$$x_{global} = x_{loc} + O_x \quad (1)$$

$$y_{global} = y_{loc} + O_y \quad (2)$$

This coordinate transformation guarantees consistent spatial alignment across slices and enables reliable aggregation of detections across the entire frame. A visualization of the SAHI slicing and stitching process is shown in Fig. 2.

D. CLIP-based Action Classification

Following human detection, the localized human regions are cropped from the original frame and passed to the action recognition module. Rather than training a conventional convolutional neural network from scratch, we employ a CLIP-based semantic classification approach that leverages large-scale pretrained knowledge.

1) *Architecture*: We adopt a linear probe / adapter-based strategy to preserve CLIP's pretrained representations. Specifically, the visual encoder of CLIP (ViT-B/32) is kept frozen during training. Each cropped human image is encoded into a 512-dimensional feature vector using the CLIP visual backbone.

On top of this embedding, we append a lightweight Multi-Layer Perceptron (MLP) adapter consisting of a 512-dimensional hidden layer followed by a ReLU activation and Dropout for regularization. The final output layer maps the features to 13 action class logits corresponding to the target action categories.

This design allows efficient adaptation to the aerial action recognition domain while avoiding overfitting and excessive computational cost.

2) *Training Objectives*: The MLP adapter is trained using the standard Cross-Entropy Loss function. Given a training sample o and class index c , the loss is defined as:

$$\mathcal{L} = - \sum_{c=1}^M y_{o,c} \log(p_{o,c}) \quad (3)$$

where $y_{o,c}$ denotes the binary indicator for the ground truth class, and $p_{o,c}$ represents the predicted probability for class c obtained via a softmax operation.

By freezing the CLIP backbone and training only the adapter, the proposed framework achieves strong generalization while maintaining training efficiency and stability.

E. Discussion

The proposed two-stage framework directly addresses the core limitations of prior aerial action recognition systems. By preserving spatial resolution during detection and leveraging semantic representations during classification, the system avoids the cascading failures commonly observed in end-to-end pipelines. Furthermore, the modular design enables independent optimization of detection and recognition components,

making the framework flexible and extensible for real-world deployment.

IV. ALGORITHM AND COMPLEXITY

A. Algorithmic Summary

Algorithm 1 Hybrid YOLO–SAHI–CLIP Aerial Human Action Recognition

Require: High-resolution aerial frame $I \in \mathbb{R}^{H \times W \times 3}$
Ensure: Action labels $\{a_1, a_2, \dots, a_N\}$ for detected humans

- 1: Initialize YOLOv11 detector D
- 2: Initialize frozen CLIP visual encoder E_v
- 3: Initialize trainable action adapter A
- 4: Initialize empty detection set $\mathcal{B}$
- 5: Compute slicing stride based on overlap ratio α
- 6: Generate overlapping slices $\{S_1, S_2, \dots, S_K\}$ from I
- 7: **for** each slice S_k **do**
- 8: Detect humans using $D(S_k)$
- 9: Map local detections to global coordinates
- 10: Add detections to $\mathcal{B}$
- 11: **end for**
- 12: Apply global Non-Maximum Suppression (NMS) on $\mathcal{B}$
- 13: **for** each bounding box $b_i \in \mathcal{B}$ **do**
- 14: Extract human crop x_i
- 15: Apply context padding and resizing
- 16: Extract CLIP embedding $z_i = E_v(x_i)$
- 17: Predict action label $a_i = A(z_i)$
- 18: **end for**
- 19: **return** $\{a_1, a_2, \dots, a_N\}$

V. EXPERIMENTS

This section describes the experimental setup, implementation details, evaluation metrics, and comparative analysis used to assess the effectiveness of the proposed hybrid framework for aerial human action recognition.

A. Implementation Details

1) *Data Preparation:* Training and evaluation were conducted on tiled image chips generated using a custom pre-processing script. Since the primary objective of the system is human action recognition, only tiles containing human instances were retained. This selective sampling strategy enables the detector to focus on recognizing humans at high spatial resolution, improving detection reliability in aerial imagery.

To ensure realism and generalization, datasets similar in structure and complexity to the *Aeriform In-Action* dataset were utilized. These datasets contain high-resolution drone footage captured under real-world conditions, including varying altitudes, illumination changes, and background clutter.

2) *Data Augmentation:* To improve robustness against scale variation and occlusion, Mosaic and MixUp augmentations were employed during training. These augmentations simulate different drone altitudes, partial occlusions, and crowd densities, thereby enhancing the model’s ability to generalize across diverse aerial surveillance scenarios.

3) *Challenges Addressed:* A major challenge in aerial action recognition is the domain shift introduced by top-down viewpoints, which significantly alters human appearance compared to ground-level imagery. To address this issue, the CLIP-based action adapter was fine-tuned to recognize humans from aerial perspectives. Additionally, few-shot domain adaptation techniques were leveraged to bridge the gap between training datasets and real-world inference conditions, improving robustness without requiring extensive retraining.

B. Results

The proposed framework achieved an overall classification accuracy of **94.53%** on the validation set. The model demonstrates exceptional performance on geometrically distinct actions such as *Lying*, *Reading*, and *Sitting*, achieving perfect precision and recall scores. More complex interaction-based actions, including *Hugging* and *Carrying*, also achieved performance above 98%, highlighting the effectiveness of semantic representations in aerial contexts.

However, the model exhibits reduced performance for the *Standing* class, with a recall of 0.65. This confusion primarily arises due to the absence of explicit temporal modeling, as individuals captured mid-stride during walking often appear visually similar to stationary standing poses in single-frame aerial imagery.

TABLE I: Drone Action Recognition: Classification Report

Action Class	Precision	Recall	F1-Score	Support
Carrying	0.99	0.99	0.99	892
Handshaking	0.94	0.95	0.94	1178
Hugging	0.99	0.98	0.98	460
Lying	1.00	1.00	1.00	701
Reading	1.00	1.00	1.00	585
Running	0.95	0.92	0.93	335
Sitting	0.99	1.00	1.00	656
Standing	0.83	0.65	0.73	596
Walking	0.91	0.95	0.93	2970
Waving	0.99	0.95	0.97	410
Macro Avg	0.96	0.94	0.95	8783
Weighted Avg	0.94	0.95	0.94	8783

C. Comparative Analysis

To further evaluate the effectiveness of the proposed hybrid architecture, we compared it against standard YOLO-based detection pipelines and video-sequence-based models such as LSTM architectures. As summarized in Table ??, the proposed approach significantly outperforms standard methods in small-object detection while requiring substantially less training data.

The integration of SAHI enables excellent detection of small-scale humans by avoiding resizing loss, whereas traditional YOLO pipelines suffer from degraded performance due to aggressive downsampling. Furthermore, the CLIP-based semantic classification module provides high contextual understanding using pretrained representations, in contrast to label-only supervision used by conventional CNN-based models.

Although the inference latency of the proposed system is moderate due to slice-based processing, it remains suitable for offline analysis and low-frame-rate surveillance applications. Importantly, the framework achieves superior performance without relying on computationally expensive temporal models, making it scalable for real-world deployment.

TABLE II: Comparative Analysis of Aerial Human Action Recognition Approaches

Feature	Our Hybrid Model	Standard YOLOv8/11	3D-CNN / LSTM
Small Object Detection	Excellent (SAHI)	Poor (Resizing loss)	N/A (Requires pre-detected crops)
Training Speed	Fast (Only adapter trained)	Medium	Slow (Large video volumes)
Context Understanding	High (CLIP semantic knowledge)	Low (Class labels only)	High (Temporal context)
Data Requirements	Low (CLIP generalizes with few samples)	High (Requires thousands of samples)	Very High (Video sequences required)
Inference Latency	Moderate (SAHI slicing overhead)	Very Fast	Slow

In addition to architectural comparisons, a quantitative performance analysis highlights the advantages of the proposed method over the base Aeriform in-action framework. The baseline approach introduced by Kapoor *et al.* employs Scaled YOLOv4 for human detection and an M-ResNeXt101 network for action recognition, achieving a test accuracy of 76.44% on the Aeriform in-action dataset. While effective for coarse action categorization, the baseline model struggles with small-scale human subjects due to aggressive image resizing and relies on a fully supervised CNN trained from scratch, resulting in limited generalization and higher data dependency.

In contrast, the proposed hybrid YOLO–SAHI–CLIP architecture achieves an overall accuracy of 94.53% on the same dataset, representing an absolute improvement of approximately 18% over the base paper. This performance gain is primarily attributed to the slicing-based inference strategy, which preserves spatial resolution for small-object detection, and the use of a CLIP-based semantic encoder that leverages large-scale multimodal pretraining. Unlike M-ResNeXt101, which requires extensive labeled data and deep feature learning, the CLIP adapter generalizes effectively under few-shot conditions and demonstrates superior performance on interaction-heavy actions such as Hugging and Carrying, where accuracies exceed 98%.

Furthermore, the proposed approach avoids computationally expensive temporal modeling used in several video-based baselines, while still outperforming them in overall classification accuracy. Although the lack of explicit temporal context leads to minor confusion between visually similar actions such

as Standing and Walking, the system remains significantly more robust and scalable for aerial surveillance scenarios compared to the baseline framework. These results confirm that decoupling detection and semantic classification provides a more effective solution for small-target action recognition in high-altitude drone imagery.

TABLE III: Quantitative Comparison with Base Aeriform In-Action Framework

Metric	Base Paper	Proposed Method
Detection Model	Scaled YOLOv4	YOLOv11 + SAHI
Action Classifier	M-ResNeXt101	CLIP (Adapter-based)
Input Resolution Handling	Full-frame resizing	Slice-based inference
Overall Accuracy (%)	76.44	94.53
Small Object Detection	Moderate	Excellent
Data Requirement	High (from-scratch training)	Low (few-shot adaptation)
Inference Suitability	Limited on edge devices	Scalable for offline / low-FPS use

In addition to comparisons with CNN-based baselines, we further evaluate our approach against recent transformer-based aerial action recognition models. In particular, we compare our framework with Drone-HAT, a hybrid attention transformer designed for complex and recurrent multi-label action recognition in drone surveillance videos. Drone-HAT employs a multi-stage pipeline involving object detection, multi-object tracking, spatiotemporal tube construction, and a multi-scale, multi-granularity vision transformer for action recognition. When evaluated on the Okutama dataset, Drone-HAT reports a Top-1 accuracy of 60.76%, representing a significant improvement over earlier 3D-CNN-based methods.

Despite its strong performance, Drone-HAT relies on explicit temporal modeling, recurrent tube generation, and transformer-based attention mechanisms, resulting in high computational complexity and increased inference latency. In contrast, our proposed YOLO–SAHI–CLIP framework achieves an accuracy of 94.53% on the Aeriform In-Action dataset without relying on temporal sequences or tracking. By decoupling detection and semantic classification and leveraging pretrained multimodal representations, our method demonstrates substantially higher robustness to small-scale subjects while maintaining lower data requirements and simpler training dynamics.

While Drone-HAT is well-suited for multi-label and recurrent action analysis, its reliance on dense temporal context limits its scalability for lightweight or offline surveillance scenarios. Our approach, although frame-based, provides superior performance for single-action recognition in ultra-high-resolution aerial imagery, highlighting the effectiveness of combining slice-based detection with semantic vision-language representations.

TABLE IV: Comparison with Drone-HAT Framework

Aspect	Drone-HAT (CVPRW 2024)	Proposed Method
Dataset	Okutama Action	Aeriform In-Action
Action Type	Multi-label, recurrent	Single-label, frame- based
Detection Strategy	YOLOv8 + tracking	YOLOv11 + SAHI
Temporal Modeling	Explicit (action tubes)	None (frame-level)
Backbone	Hybrid Attention Transformer	CLIP (ViT-B/32) Adapter
Overall Accuracy (%)	60.76	94.53
Small Object Robustness	Moderate	High
Computational Complexity	High	Moderate

VI. CONCLUSION

This research demonstrates that separating aerial human action recognition into two specialized components—human detection and semantic action classification—provides a robust and scalable solution for small-target recognition in drone-based surveillance. By explicitly decoupling these tasks, the proposed framework avoids the cascading failures commonly observed in end-to-end pipelines when applied to high-altitude aerial imagery.

The integration of YOLOv11 with Slicing Aided Hyper Inference (SAHI) enables precise localization of small-scale human subjects by preserving spatial resolution during inference. In parallel, the use of a CLIP-based semantic classification module leverages large-scale multimodal pretraining to disambiguate visually similar actions observed from top-down perspectives. This combination of mechanical precision and semantic reasoning allows the system to achieve an overall accuracy of approximately 94% on challenging aerial datasets, significantly outperforming conventional CNN-based and video-sequence-based approaches.

Experimental results and comparative analysis confirm that the proposed hybrid architecture offers superior small-object detection capability, reduced data requirements, and efficient training compared to standard YOLO pipelines and computationally intensive 3D-CNN or LSTM-based models. While the absence of explicit temporal modeling leads to minor confusion between visually similar actions such as standing and walking, the framework remains well-suited for real-world surveillance applications where robustness and scalability are critical.

Future work will focus on incorporating lightweight temporal modules, such as Temporal Shift Modules (TSM), to capture short-term motion cues and resolve remaining ambiguities without sacrificing inference speed or increasing computational complexity. Overall, this work establishes a strong foundation for reliable and efficient aerial human action recognition in real-world drone surveillance scenarios.

REFERENCES

- [1] S. Kapoor, A. Sharma, A. Verma, and S. Singh, “Aeriform in-action: A novel dataset for human action recognition in aerial videos,” *Pattern Recognition*, vol. 140, p. 109505, Mar. 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0031320323002054>
- [2] M. Elnady and H. E. Abdelmunim, “A novel YOLO–LSTM approach for enhanced human action recognition in video sequences,” May 2025.
- [3] M. Khan, J. Ahmad, A. E. Saddik, W. Gueaieb, G. De Masi, and F. Karay, “Drone-HAT: Hybrid attention transformer for complex action recognition in drone surveillance videos,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 4713–4722.
- [4] S. Chen, Y. Liu, H. Zhang, and Y. Cai, “A human location and action recognition method based on improved YOLOv11 model,” Aug. 2025.
- [5] M. Wang, J. Xing, J. Mei, Y. Liu, and Y. Jiang, “ActionCLIP: Adapting language–image pretrained models for video action recognition,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2023. [Online]. Available: <https://ieeexplore.ieee.org/document/10323592>
- [6] H. Mliki, F. Bouhlel, and M. Hammami, “Human activity recognition from UAV captured video sequences,” *Pattern Recognition*, vol. 100, p. 107140, 2020.
- [7] M. Yaseen and H. Najjaran, “What is YOLOv8: An in-depth exploration of the internal features of the next-generation object detector,” arXiv preprint, 2024.
- [8] X. Liang, J. Xiang, S. Qin, *et al.*, “Small target detection algorithm based on SAHI-Improved-YOLOv8 for UAV imagery: A case study of tree pit detection,” *Smart Agricultural Technology*, 2025.
- [9] J. Redmon and A. Farhadi, “YOLOv3: An incremental improvement,” *arXiv preprint arXiv:1804.02767*, 2018.
- [10] A. M. Algamdi, V. Sanchez, and C.-T. Li, “DroneCaps: Recognition of human actions in drone videos using capsule networks with binary volume comparisons,” in *Proc. IEEE Int. Conf. on Image Processing (ICIP)*, 2022, pp. 3174–3178.
- [11] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lučić, and C. Schmid, “ViViT: A video vision transformer,” in *Proc. IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, Montreal, BC, Canada, Oct. 2021, pp. 6836–6846.